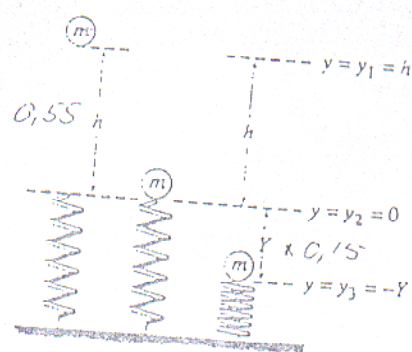


- (f) 'n Bal met massa $m = 2,60$ kg val vertikaal afwaarts vanuit rus deur 'n hoogte $h = 55$ cm. Dit val op 'n veer wat dan $Y = 15$ cm saamdruk.

A ball with mass $m = 2,60$ kg, starting from rest, falls a vertical distance $h = 55$ cm before striking a coiled spring, which compresses an amount $Y = 15$ cm.



- (i) Toon met behoud van energie aan dat die snelheid v_2 van die bal net wanneer dit die veer tref (by posisie $y_2 = 0$) is $3,28$ m/s.

From the conservation of energy show that the velocity v_2 of the ball just as it touches the spring (at position $y_2 = 0$) is $3,28$ m/s.

$$E_{\text{voor}} = E_{\text{ander}} \quad (\text{Behoud van } E)$$

$$mgh + \frac{1}{2}mv^2 = mgy_2 + \frac{1}{2}mv_2^2$$

$$16,014 = 0 + \frac{1}{2}(2,6)(v_2)^2$$

$$v_2^2 = 12,78$$

$$v_2 = 3,28 \text{ m/s}$$

(4)

- (ii) Bepaal die veerkonstante k indien die veer ten volle saamgedruk word tot, by $y_3 = -Y = -0,15$ m

Determine the spring constant k if the spring is fully compressed at $y_3 = -Y = -0,15$ m

$$F = -kx$$

$$E_1 = E_2$$

$$mgy_2 + \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = mgy_3 + \frac{1}{2}mv_3^2 + \frac{1}{2}kx_3^2$$

$$0 + \frac{1}{2}(2,6)(3,28)^2 + 0 = 2,6(9,8)(-0,15) + 0 + \frac{1}{2}k(0,15)^2$$

$$13,98592 = -3,022 + 0,1125k$$

$$k = 1582,93 \text{ N/m}$$

(5)